

Changes in Matter

Understanding Phase Changes, the Role of Thermal Energy, and Matter Conservation



Learning Competencies & Let's Explore

Learning Competencies

At the end of the lesson, learners should be able to:

- ✓ Use diagrams or flowcharts to model melting, evaporation, freezing, and condensation.
- ✓ Explain the role of heat in changes of state.

Let's Explore

Have you ever seen ice turn into water? What happens to wet clothes when they are left under the sun?

Key Idea: These are examples of changes in the state of matter. Matter can change from one state to another when energy, such as heat, is transferred.

What Are Changes in Matter?

Matter can exist as a **solid**, **liquid**, or **gas**. When matter changes from one state to another, its state changes, but the matter does not simply disappear.

Change	From (Initial State)	To (Final State)
Melting	Solid	Liquid
Evaporation	Liquid	Gas
Freezing	Liquid	Solid
Condensation	Gas	Liquid



These four state changes occur whenever thermal energy is transferred to or from matter.

Melting and Evaporation



MELTING

Solid → Liquid

Melting happens when a solid **gains heat** and changes into a liquid.

Example: Ice → Water

The ice absorbs heat from its surroundings and changes from solid to liquid.



EVAPORATION

Liquid → Gas

Evaporation happens when a liquid **gains energy** and changes into a gas.

Example: Water → Water vapor

Wet clothes become dry because some of the water changes into water vapor.

Freezing and Condensation



FREEZING

Liquid → Solid

Freezing happens when a liquid **loses heat** and changes into a solid.

Example: Water → Ice

When water is cooled enough, it changes into solid ice.



CONDENSATION

Gas → Liquid

Condensation happens when a gas **loses heat** and changes into a liquid.

Example: Water vapor → Water droplets

Water droplets form on the outside of a cold glass because vapor in the air cools and condenses.

The Role of Heat

The Role of Heat

Heat or energy transfer can cause matter to change state.

Heating may cause:

- Melting (Solid → Liquid)
- Evaporation (Liquid → Gas)

Cooling may cause:

- Freezing (Liquid → Solid)
- Condensation (Gas → Liquid)

Does Matter Disappear?

SCIENTIFIC FACT

NO! Matter does not disappear.

During a change of state, the **amount of matter remains the same.**



Concrete Example

For example, when **ice melts**, the ice does not disappear.

It simply changes into liquid water. The total mass and amount of matter stay identical!



Key Takeaway: State changes affect physical form and particle arrangement—not the existence of matter!

SOLID

LIQUID

GAS

State Change Flowchart

For each situation, identify: **Starting State** → **Ending State** → **Name of Change** → **Heat Gained or Lost**

1. Ice becomes water

Solid → _____

Heat Gained

2. Water becomes vapor

Liquid → _____

Heat Gained

3. Water becomes ice

Liquid → _____

Heat Lost

4. Vapor becomes droplets

Gas → _____

Heat Lost

Assessment (Questions 1 – 3)

QUESTION 1

What happens during melting?

- A. Liquid → Solid
- B. solid → Liquid
- C. Gas → Liquid
- D. Liquid → Gas

QUESTION 2

What role does heat play in changes of state?

- a. It can cause matter to change state
- B. Makes matter disappear
- C. Removes all matter
- D. Changes matter into nothing

QUESTION 3

What happens during condensation?

- A. Solid → Liquid
- B. Liquid → Gas
- C, Gas → Liquid
- D. Liquid → Solid

Assessment (Questions 4 – 5)

QUESTION 4

Does matter disappear when it changes state?

- A. Yes
- B. No
- C. Only when heated
- D. Only when frozen

QUESTION 5

Why can a flowchart be used to show changes of state?

- A. To show and communicate the sequence of changes
- B. To remove matter
- C. To measure temperature only
- D. To classify living things

